Aleks Hughes

London | last revised 25th March 2025

Full Stack Software Engineer with over 10 years experience. Proven high performer, built a range of complex systems including LLM-powered applications and data-intensive analytics. Combines strong technical skills, excellent communication and business understanding to deliver impactful results.

Experience

Full Stack Software Engineer | Schroders AI & Innovation | Sept 2023 to Present

- Principal developer on "Genie" an internal ChatGPT-like chatbot achieving ~3,000 weekly active users (50% of the firm). Built with python FastAPI backend, TypeScript React frontend and hosted on kubernetes on AWS with Azure integrations. Principal developer in a team of 14.
  - Built test pyramid of backend pytest/behave, frontend jest and end-to-end playwright tests.
  - Developed a range of Retrieval Augmented Generation (RAG) strategies with performance evaluations.
  - Evaluated different vector databases: in-memory, Chroma, pSQL pgVector and Azure AI Search.
  - Wrote a bulk RAG interface that enables users to repeatedly ask many questions of many documents.
  - RAG over internet search results: routing user queries that are unlikely to be answerable by the unenhanced-LLM to a search engine tool and synthesising a response using top-N results.
  - Added user-defined system instructions, settings and data sources like OpenAI's "custom GPTs".
  - Added Microsoft Graph API integrations to OneDrive and Teams; executed by LLM agentic function calls.
  - Defined a spec and execution engine for custom user-defined workflow templates. Enables users to write their own prompt templates with sequential and parallel LLM calls.
  - Added application telemetry using OpenTelemetry and grafana to capture user traces, linking traces to log entries and custom usage metrics.
  - Provided consulting to business teams to encourage adoption and enable AI-powered workflows.
  - Published AI-related APIs for shared consumption across the business.
- Owned a set of hosted jupyter notebooks demonstrating common patterns for interacting with LLMs.
- Chaired a fortnightly developer clinic to answer queries, update on latest trends and share best practice
- Experiments: using LLMs to query SQL databases, benchmarking best method to extract tables from PDFs.
- Modernising the local stack and CI pipeline from local interpreters to docker compose.

Full Stack Software Engineer | Schroders Investment Technology | Sept 2020 - Sept 2023

- Lead developer on "ActiveIQ" a web app to track an asset manager's interactions with held entities; like an ESG performance improvement plan for companies. Built with python FastAPI backend, TypeScript React frontend and hosted on AWS. Integrated with asset holdings, employee data, external reporting. CronJob scheduled actions. Custom short subdomain.
- Developer on "Country Context" a model and web application for comparing country level ESG performance. Built with Python-Django, vanilla JavaScript and pSQL. Led stakeholder comms and baselined requirements. Productionised a PoC application. Drafted architecture diagram. SSO auth. Established first production support model on the app's cloud-hosted platform.
- Lead Developer on "Thesis Capture" a web application to help investment analysts track projections versus actuals. Built with Python-Django, JS and pSQL. Hosted on AWS. Value-investing desk integration: customised the app for their workflow. Designed data model for 5-year backfill of 3,000+ Excel valuations, used data versioning. Built integrations with Denodo, QAI and other external data providers.
- Developer, SustainEx: python model for company and fund-level Article 8 and 9 ESG performance. System architecture and design. Use case and requirement hardening. Wireframed and built frontend analytics flow.
- Python Appreciation Society: promoting python best practice. Founding member and first session speaker.
- Recruitment and interviewer: hired 8 engineers. Developed a submission scorecard with categories for code readability, algorithm design, language internals.
- Hackathon team competition winner: built an app to automate infrastructure catalogue updates. Wrote the application's TOML file parsing logic, iServer API and email interface.

Consultant Data Scientist | Parker Fitzgerald (Acquired by Accenture) | Apr 2017 - Sept 2020

- Developed Python visualization tools for HSBC to detect mis-selling. Used unsupervised learning to identify suspicious patterns among millions of trades. Contributor to the Bank's strategic python platform 'StratPy'.

- Created unsupervised model for Open Banking fintech that segmented 50M transactions into 400 categories, improving suspicious activity detection by 10%.
- Developed ARIMA-X predictive model for Lloyds to identify higher risk business lines, successfully forecasting 4 high-impact events during backtesting.
- Built an R-Shiny application for Lloyds that standardized data sampling methods, winning "best use of technology" industry recognition.
- Developer on NLP application for UBS to classify change ticket documentation and provide early indicators of technology change risk.

Volunteer NLP Engineer | Full Fact | Oct 2018 - Feb 2019
- Project to 'automate fact checking' through NLP. Project led by a core maintainer of NLP package 'genism'.
- Developed models to parse a claim for geography, direction and time frame.
- Classification models displayed average 40% precision uplift over existing rule-based heuristic models.

Technical Consultant | IBM | Oct 2014 - Apr 2017
- Graduate Managers Choice Annual Award 2016. IBM Europe Award 'Putting Client First' August 2015.
- Python Developer for data remediation at Deutsche Bank. Scripted multi-threaded ETL pipeline.
- Python Developer on data migration at Barclaycard. Wrote logic to map 300,000+ end customers to 1,500+ target billing and payment configurations.
- Organised 17 Lunch and Learns, topics include DevOps, Design Thinking, Agile and Blockchain.

Skills
Technical
- Advanced Python: FastAPI, django, pytest, pandas, numpy, scikit-learn, langchain.
  Async, multiprocessing, decorators, generators, high performance.
- Fluent: JavaScript/TypeScript, React, Jest.
- Good knowledge: Java/Kotlin, R, Rust.
- Computer Science: Algorithms, data structures, distributed systems.
- Software Design: OOP, functional programming, test-driven development.
- Cloud & DevOps: AWS and Azure, Docker, Kubernetes, CI/CD pipelines.

Non-technical
- Native English and fluent Spanish.
- Team leader: writing team manifesto documents, conducting PR reviews, architecture decisions.
- Business analysis and requirement gathering: understanding and documenting 'as-is' and 'to-be'.
- Product demos and non-technical communications.

Interests
Keen cyclist and swimmer. University Basketball 2nd team. 6 months volunteer English teacher in rural Mexican primary school. Combined 10 weeks Spanish immersion in Cuba and Spain. Biked 4,500Km through Alaska, hiked in the Pyrenees, Patagonia and Atlas Mountains and dogsledding in Yukon. Volunteer A-level Maths and Economics tutor to underprivileged kids, theaccessproject.org.uk.

Education
MSc Economics and Econometrics | Trinity College Dublin | August 2014
- Thesis title: 'How do Oil Price Shocks Affect Scottish Real Economic Activity?'
- Selected for the Trinity College's 'Econometrics Game' team 2014, econometricgame.com.

BSc Hons Economics and Econometrics 2.1 | University of Nottingham | August 2012
- Writer for student run newsletter 'Nottingham Economic Review', neronline.co.uk.

A-levels | St Peters School, York | July 2008
- Maths (A), Further Maths (A), Economics (A), History (B) and Politics (A, AS-level).
- UK Maths Trust Olympics team. Silver in UKMT Senior Challenge. Vice-Captain of Rowing.